

AdreamDCO: AI-Driven Robust and Efficient Design Automation for Digitally Controlled Oscillators

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Abstract—This paper presents how we leverage AI-human collaboration to develop an end-to-end, automated design flow for digitally controlled oscillators (DCOs), a key radio-frequency (RF) integrated circuits (ICs) building block that dominates phase noise and jitter performance of RF systems. Specifically, we decompose the DCO design process into two steps and use AI to enhance productivity and optimize performance within each step. Additionally, we demonstrate how AI can assist RF IC designers in creating unconventional circuit components to tackle challenging design specifications. Overall, the proposed flow is capable of synthesizing the DCO design including the schematic and layout in 80 seconds after one-time training, and is frequency agile between 1 and 20 GHz. Moreover, it can select the most robust design under process variations when multiple design parameters meet target specifications under the nominal condition. The proposed automated DCO design flow is demonstrated using two silicon prototypes implemented in the GlobalFoundries 22-nm CMOS SOI process. In the measurements, they achieve >192.4 -dBc/Hz figure-of-merit (FoM) and <1.5 -kHz frequency resolution at 7.1 to 8.6 GHz and 3.8 to 4.6 GHz, outperforming existing manual designs at similar frequencies.

Index Terms—artificial intelligence (AI), Bayesian optimization (BO), CMOS, design automation, differential evolution (DE), DCO, frequency tuning, RF IC, residual neural networks.

I. INTRODUCTION

RF ICs are foundational to wireless communication and sensing systems, enabling applications such as Wi-Fi connectivity, cellular communications, automotive radars, and many more. Similar to analog ICs, the existing RF IC design flow is mostly manual and heavily dependent on human expertise. As the demand for higher performance and increased complexity in RF systems continues to grow, the conventional manual design flow poses significant challenges in meeting the target product-to-market time and performance specifications. Additionally, the current manual design flow involves extensive electromagnetic (EM) and nonlinear circuit simulations, both of which are computationally intensive, leading to long simulation times. Reliability issues, such as process variations, further complicate RF IC design and verification. Consequently, RF IC design automation is emerging as an active area of research [1]–[8].

Several recent works that leveraged AI to automate the design of passive networks within RF ICs have been reported [9]–[18]. While these tools can greatly reduce the design

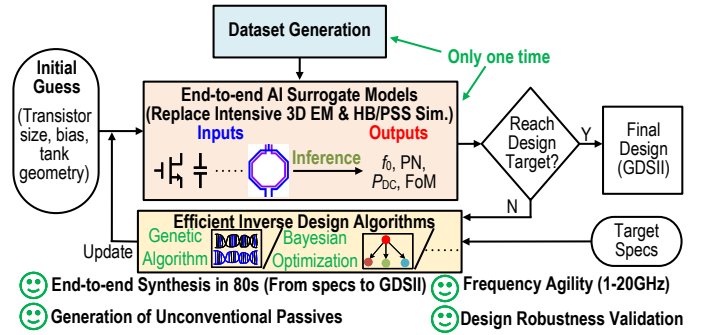


Fig. 1. Our proposed AI-assisted, automated DCO design flow and its key features.

time for passive networks, it is important to note that the design of active circuits (i.e., building block topology, transistor sizing, and biasing) is equally crucial as the design of passive networks for RF ICs. Even after the initial passive network is ready, its integration with active components still requires extensive nonlinear circuit simulations and further optimizations to account for transistor nonidealities (such as parasitics and nonlinearity). This highlights the need for an “end-to-end” design automation flow that can jointly optimize active and passive components to streamline the overall RF IC design process.

To address this unmet need, this paper presents an AI-assisted end-to-end design flow and uses an LC-based digitally controlled oscillator (DCO) as a demonstration example. The LC-based DCO is one of the most critical building blocks in modern wireless and wireline transceivers, and its performance dominates the overall phase noise and jitter of the system clock. As shown in Fig. 1, our AI-assisted design flow consists of three steps: (i) generating datasets (as a one-time effort); (ii) training high-fidelity, end-to-end AI surrogate models to replace computationally intensive 3-D EM and nonlinear circuit simulations, enabling rapid evaluation of DCO performance based on active and passive design parameters; and (iii) developing efficient inverse design algorithms to automatically find the optimal design.

Our major contributions are summarized as follows:

- **End-to-end Synthesis for Active and Passive Co-**

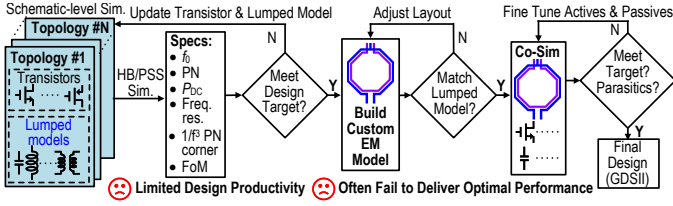


Fig. 2. Conventional manual DCO design flow.

Design: Once trained, the GDSII of the DCO tank, along with design parameters of the DCO transistor core (with the consideration of layout parasitics), is directly synthesized from target specifications within 80 seconds.

- **Generation of Unconventional Passives:** We demonstrate that the automated flow is capable of assisting human designers in creating new, unseen passive networks that outperform existing expert designs.
- **Frequency Agility:** The proposed flow can produce designs across a wide frequency range, anywhere between 1 and 20 GHz.
- **Design Robustness Validation:** The AI model enables rapid process variation simulations of the DCO tank, a task that is challenging using conventional EM solvers.

II. BACKGROUND

A. Conventional, Manual DCO Design Flow

The conventional design flow of DCOs is both time-consuming and labor-intensive. As shown in Fig. 2, it begins with selecting a suitable DCO topology based on target specifications, a process that often requires deep design expertise. Next, designers build the DCO schematic and perform schematic-level simulations. In this step, lumped-element models are usually constructed to model the passive components in the circuit [e.g., inductors and transformers (XFMRs) [19]–[21]]. Despite their limited accuracy compared with scattering parameters (S-parameters) derived from 3-D EM simulations, these lumped-element models are crucial in manual designs, as they provide valuable intuitions for human designers and can guide subsequent optimizations.

Next, the transistor design parameters and lumped-element models are iteratively tuned to meet target DCO specifications. Typical specifications include oscillation frequency (f_0), phase noise (PN), DC power consumption (P_{DC}), frequency resolution, $1/f^3$ phase noise corner, and figure-of-merit (FoM), as shown in Fig. 2. Among these, the phase noise and $1/f^3$ phase noise corner are related to the frequency stability of the DCO output, which serves as the clock signal of the RF system. In this work, we monitor both PN_{10MHz} to evaluate far-end phase noise at 10-MHz offset and PN_{10kHz} to assess close-in phase noise at 10-kHz offset. A smaller difference between PN_{10kHz} and PN_{10MHz} indicates a better (lower) $1/f^3$ phase noise corner. Additionally, a low P_{DC} is highly desired, especially for battery-powered devices to extend battery life. Fine DCO frequency resolution is also preferred in communication systems for precise channel selection. Finally, the FoM

is a comprehensive benchmark that takes f_0 , phase noise, and P_{DC} into consideration. DCO design always aims to maximize FoM while ensuring all other specifications meet the system requirements.

After the schematic-level design, 3-D EM models of the passive components are created. Designers manually adjust the geometric parameters of the EM models, run EM simulations, and extract circuit parameters from the simulation results. This step is repeated until the extracted circuit parameters match those of the lumped-element model.

Following this step, the EM-simulated S-parameters of the passive components are imported into the circuit design environment for co-simulation with the active components. Given that RF IC performance is highly sensitive to device parasitics and nonlinearity, designers typically need to fine-tune the geometry of the 3-D EM models again and jointly optimize them with the active part until the overall DCO achieves the target performance.

This iterative and heuristic process is not only time-consuming but also prone to sub-optimal results, as the large design space increases the risk of being trapped in local optima. Moreover, discrepancies often arise between the lumped-element model and EM-simulated S-parameters of the passive components, especially for wide-band designs. Consequently, design productivity and overall performance are often limited by this manual design process.

B. Recent Advances in RF IC Design Automation

A few works have recently been reported that leveraged AI surrogate models to automate the design of *passive networks* within RF ICs. These models include convolutional neural networks (CNN) [9], multilayer perceptrons (MLP) [10], recurrent neural networks (RNN) [11], graph transformers [12], UNets [13], and graph neural networks (GNN) [14]. They are capable of predicting either the S-parameters of passive networks [9], [11]–[14] or their electrical parameters [10], [15]–[17], such as input impedance and loss, presented to active components. Beyond template-based synthesis, AI models have also been applied to non-heuristic passive network designs, such as the pixelated network in [18] and the XFMR with localized deformation in [13]. However, as mentioned earlier, *what is currently missing is the co-design of active and passive components in an end-to-end manner*. This gap is specifically addressed in our proposed design flow.

III. PROPOSED AI-ASSISTED DCO DESIGN FLOW

A. Decomposing DCO Design Into Two Steps

As demonstrated in recent advancements in AI-assisted analog and RF IC design automation [2], [10], [14], [16], [18], adopting “AI-human collaboration” represents the optimal strategy, as it offers explainable and trustworthy designs while harnessing the full potential of AI. More importantly, *it allows circuit designers to focus on establishing circuit insights and exploring new circuit ideas without being bogged down by nuances*.

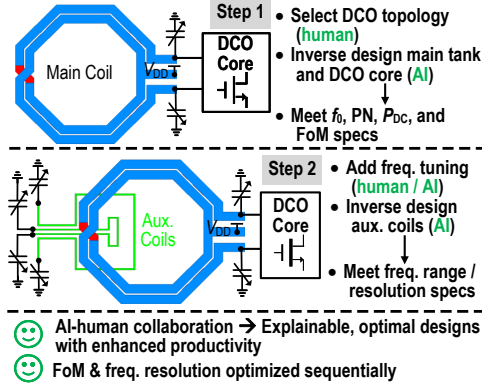


Fig. 3. Our proposed two-step DCO design automation flow.

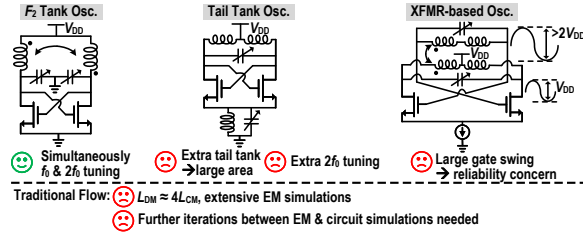


Fig. 4. Comparison between different main-tank topologies of the oscillator.

As shown in Fig. 3, we leverage this AI-human collaboration to guide our development and decompose the DCO design process into two steps. In the first step, we design the main tank and DCO transistor core to achieve the target phase noise and FoM. The second step then focuses on optimizing frequency tuning to meet the desired frequency resolution. The importance of this two-step approach is that the two most critical DCO metrics, i.e., FoM and frequency resolution, can be optimized sequentially. This significantly simplifies our development efforts, making it possible to establish an efficient automated design flow.

Within each step, we either rely on human expertise to select an existing topology (such as in *step 1*) or collaborate AI to develop a new topology (such as in *step 2*). Additionally, AI is employed in both steps to assist designers in enhancing design productivity and optimizing circuit performance.

B. Step 1: DCO Main Tank Design

There are three dominant oscillator topologies that present high FoM (see Fig. 4): the Class- F_2 tank topology [22]–[24], the tail-tank topology [25], [26], and XFMR-based topology [27], [28]. In this work, we selected the Class- F_2 topology for the design of the main tank and DCO core, as it achieves simultaneous fundamental (f_0) and second harmonic ($2f_0$) tuning without the need for a dedicated tail tank, while ensuring more reliable operation compared with the XFMR-feedback topology. However, the Class- F_2 tank requires the L_{DM} (differential-mode inductance) / L_{CM} (common-mode inductance) ratio to be ≈ 4 for simultaneous f_0 and $2f_0$ tuning [22]. In the traditional manual design flow, optimizing

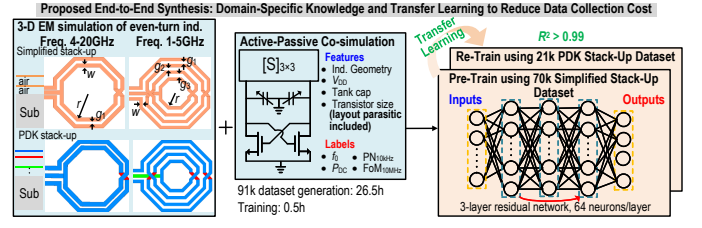


Fig. 5. Dataset generation and end-to-end AI surrogate model training of DCO main tank design.

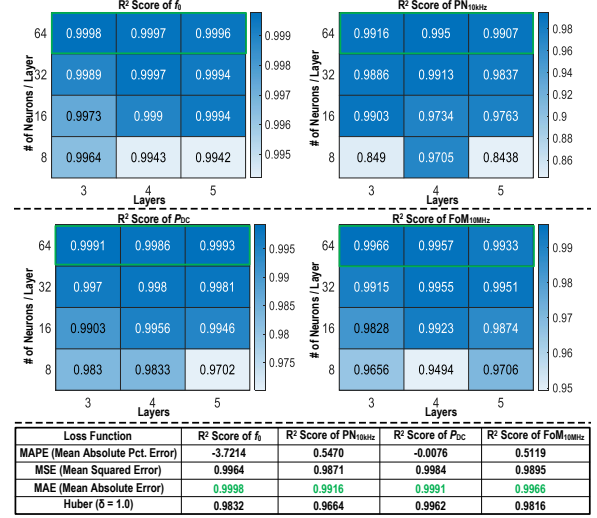


Fig. 6. Comparison between different NN hyperparameters and loss functions.

such an inductor requires extensive EM simulations. Even after the initial inductor layout is ready, further iterations between passive and active co-optimizations are still needed to achieve the best possible FoM.

Our automated design flow addresses these challenges in an end-to-end manner, eliminating the need to explicitly consider the L_{DM} / L_{CM} ratio as an intermediate step. To reduce the data collection overhead associated with training high-fidelity models, we leveraged domain-specific knowledge and transfer learning (TL) [18]. For example, we focused on inductors with an even number of turns (2-turn and 4-turn inductors in this work to cover 1 to 20 GHz), as they naturally offer a smaller L_{CM} compared with inductors with an odd number of turns [22], making it easier to satisfy the desired $L_{DM} / L_{CM} \approx 4$ relationship. Additionally, we first collected datasets using a simplified stack-up to train an initial model, which was then fine-tuned with a smaller number of samples collected using the complete PDK-specific stack-up, as shown in Fig. 5. This TL approach significantly reduces the dataset collection time by $\sim 3\times$. Once the S-parameters of the inductors were obtained from EM simulations, an end-to-end dataset was generated by co-simulating the S-parameters and the DCO core in Cadence Virtuoso. The overall data collection took ~ 26.5 hours on a 128-core workstation, generating $\sim 91k$ data points.

We used the residual network architecture to build our AI

TABLE I
DESIGN EXAMPLES COVERING 1 TO 20 GHz

Frequency	1.6 GHz	4.0 GHz	8.0 GHz	10.0 GHz	12.0 GHz	14.0 GHz	16.0 GHz	20.0 GHz
Main Tank Layout								
Transistor Size	200μm/100nm	40μm/100nm	40μm/100nm	40μm/100nm	20μm/100nm	40μm/100nm	40μm/100nm	60μm/100nm
SE Cap	1400 fF	702 fF	654 fF	443 fF	257 fF	226 fF	207 fF	205 fF
FoM _{10MHz}	193.5 dBc/Hz	193.1 dBc/Hz	193.5 dBc/Hz	193.4 dBc/Hz	194.0 dBc/Hz	192.5 dBc/Hz	192.1 dBc/Hz	189.6 dBc/Hz
P _{DC}	0.78 mW	0.51 mW	0.66 mW	0.65 mW	0.38 mW	0.66 mW	0.71 mW	1.14 mW

TABLE II
COMPARISON BETWEEN DE AND BO

Algorithm	DE	BO
SE Capacitor Value (fF)	654	992
Inductor Radius (μm)	80	56
Inductor Coil Gap (μm)	3	14
Inductor Coil Width (μm)	14	8
Transistor Size	40μm / 100nm	40μm / 100nm
f ₀ (GHz)	7.9996	8.0820
PN _{10kHz} (dBc/Hz)	-73.9282	-59.0346
P _{DC} (mW)	0.664	0.827
FoM _{10MHz} (dBc/Hz)	193.51	192.39
Elapsed Time (seconds)	36.9	33.5

surrogate model, due to its superior accuracy and scalability [15], [29]. During training, we experimented with different hyperparameters and loss functions, as shown in Fig. 6. We eventually chose 3 layers, 64 neurons per layer, and Mean Absolute Error (MAE) as the loss function, balancing between training overhead and accuracy. The model is lightweight, thanks to the domain-specific knowledge and transfer learning that were applied during data collection. The predicted R^2 score exceeds 0.99.

Once the surrogate model was ready for performance evaluation, we solved the specifications-to-parameters (S2P) inverse design problem using optimizations. Specifically, the inverse design problem was formulated as

$$\begin{aligned} \arg \min_{\Phi} \quad & -\text{FoM}_{10\text{MHz}} \\ \text{s.t.} \quad & |f_0 - f_{\text{target}}| = 0, \quad \text{PN}_{10\text{kHz}} < \text{PN}_{10\text{kHz},\text{target}}, \\ & P_{\text{DC}} < P_{\text{DC},\text{target}} \end{aligned} \quad (1)$$

where Φ represents the design space for both active and passive components, and $\text{FoM}_{10\text{MHz}}$ is the optimization object. The constraints include f_0 , $\text{PN}_{10\text{kHz}}$, and P_{DC} .

A penalty function $\Psi(\Phi)$ was introduced, as

$$\begin{aligned} \Psi(\Phi) = & -\omega_1 \times \text{FoM}_{10\text{MHz}} + \omega_2 \times |f_0 - f_{\text{target}}| \\ & + \omega_3 \times \text{elu}(\text{PN}_{10\text{kHz}} - \text{PN}_{10\text{kHz},\text{target}}) \\ & + \omega_4 \times \text{elu}(P_{\text{DC}} - P_{\text{DC},\text{target}}) \end{aligned} \quad (2)$$

where $\omega_{1,2,3,4}$ are weights assigned to respective specifications, and the exponential linear unit (*elu*) penalizes constraint violations [30].

It is important to note that in the DCO design, the cross-coupled active core must provide sufficient loop gain to sustain

oscillation. To exclude design parameters that fail to sustain oscillations, we trained an additional three-layer neural network to pre-screen design parameters, and any design failing to oscillate was assigned a large penalty.

We explored two methods to solve this inverse design problem: the differential evolution (DE) algorithm and Bayesian optimization (BO). To compare the performance between the two, we ran an experiment with the following design targets: $f_0 = 8.0$ GHz, $\text{PN}_{10\text{kHz}} < -75$ dBc/Hz, and $P_{\text{DC}} < 0.7$ mW. As shown in Table II, both DE and BO generated designs within 40 seconds. Additionally, the design synthesized by DE outperformed that by BO, particularly in terms of f_0 accuracy, $\text{PN}_{10\text{kHz}}$, P_{DC} , and $\text{FoM}_{10\text{MHz}}$. This advantage arises because DE directly utilizes the AI surrogate model to predict DCO performance, enabling nearly instantaneous performance evaluations. In contrast, BO relies on a Gaussian process regressor (GPR) to predict the objective function output. The GPR is iteratively updated using data points generated by the AI surrogate model, resulting in slower evaluations, reduced accuracy, and less effective convergence. Based on this experiment, we chose DE to solve the rest of the inverse design problems in this work.

Our AI surrogate model and inverse design are applicable anywhere between 1 and 20 GHz. Table I summarizes the inverse design results for the main tank and its associated single-ended (SE) capacitance and transistor size at a few frequencies across the 1–20 GHz range. All these designs achieved FoM of >189 dBc/Hz. Additionally, their performance predicted by the surrogate model closely matches the results obtained from EM and nonlinear circuit simulations, demonstrating the effectiveness of our approach.

C. Step 2: Adding Frequency Tuning Mechanism to the Main Tank

As for the second step of the DCO design, we demonstrate how we leverage AI to assist with the design of a new EM structure for fine frequency tuning. In this work, we set a challenging frequency resolution target of sub-1.5 kHz. This requires a switched capacitor array (SCA) with a total of 21 bits, assuming a typical 20% tuning range with margin. To enable design automation, we chose to use existing PDK capacitors instead of manually designing custom capacitors as in [26], [28], as this avoids the extensive overhead associated

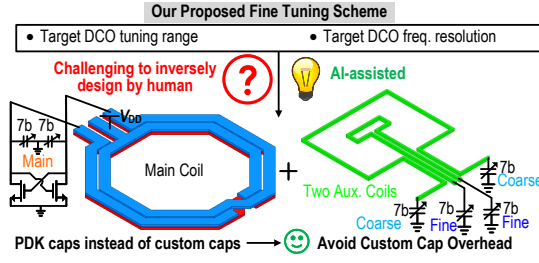


Fig. 7. Our proposed fine frequency tuning scheme incorporating an unconventional EM structure co-developed by AI.

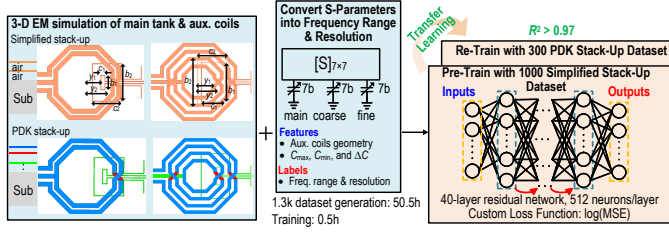


Fig. 8. End-to-end surrogate model for fine frequency tuning.

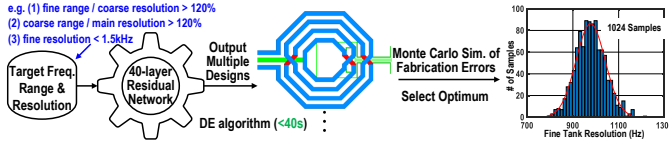


Fig. 9. AI-assisted design automation to identify the optimal layout of the auxiliary coils, achieving the target sub-1.5-kHz frequency resolution.

with manual layout, parasitic extraction, EM simulations, fine tuning, and potential mismatches between custom capacitor models and actual measurements.

Past work has shown that when an auxiliary coil is loaded with a capacitor and weakly coupled to the main tank, the effective capacitance presented to the main tank is significantly smaller [28], [31]. However, a single auxiliary coil is insufficient to meet our stringent frequency resolution target of sub-1.5 kHz, as the minimum capacitor required for a single auxiliary coil is still impractically small. As shown in Fig. 7, our design intuition suggests that introducing one more auxiliary coil, i.e., two auxiliary coils in total, with different magnetic couplings to the main tank could address this problem. While this high-level circuit idea is promising, deriving closed-form equations to guide the design is overly complex. Moreover, in the traditional manual design flow, we must rely on a lumped-element model with numerous mutually coupled parameters for this new EM structure to complete the initial schematic-level design. Optimizing this lumped-element model to achieve the desired frequency tuning range and resolution is itself challenging, let alone finding an EM structure that aligns the lumped-element model. All these intricate design and layout nuances necessitate collaboration with AI.

As shown in Fig. 8, similar to the *Step 1*, we began by generating datasets with different geometric parameters of the

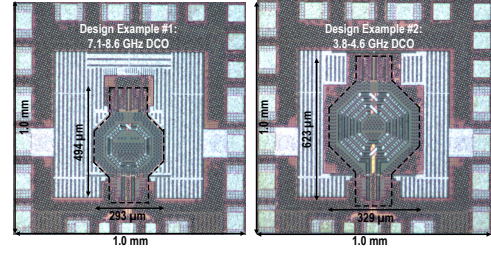


Fig. 10. Die micrographs of the two DCO designs implemented in the GlobalFoundries 22-nm CMOS SOI process.

auxiliary coils and recording their S-parameters from EM simulations. These S-parameters were then co-simulated with the main-coil SCA, coarse-tuning SCA, and fine-tuning SCA to derive the frequency tuning range and resolution for each SCA: f_{TR1} and f_{Res1} for the main coil, f_{TR2} and f_{Res2} for the coarse-tuning coil, and f_{TR3} and f_{Res3} for the fine-tuning coil. The $f_{TR1,2,3}$ and $f_{Res1,2,3}$ are further saved as “labels” for training.

We applied the same TL approach as in Section III-B to reduce data collection overhead. A total of 1,000 passive structures were simulated using a simplified metal stack, and an additional 300 samples were simulated with the PDK-specific metal stack. Next, a 40-layer AI surrogate model based on the residual architecture was trained. The gradients of traditional MSE and MAE tend to diminish when their values are very small, which significantly slows down the convergence. To address this issue, we introduced a logarithmic term in the loss function, as

$$loss = \log_{10} \left[\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \right] \quad (3)$$

where Y_i is the ground-truth label, and $\hat{Y}_i$ is the predicted value. This loss function enabled faster convergence, and the model achieved an R^2 score of >0.97 during testing.

Finally, the inverse design was performed using DE. Similar to Section III-B, the inverse design problem was formulated as an optimization problem, as

$$\arg \min_{\Xi} f_{Res3}, \quad \text{s.t.} \quad \frac{f_{TR2}}{f_{Res1}} > \delta_1, \quad \frac{f_{TR3}}{f_{Res2}} > \delta_2 \quad (4)$$

where Ξ represents the layout parameters of the auxiliary coils, and the constraints are to ensure that the tuning range of the coarse (or fine) SCA is at least $\delta_1 \times$ (or $\delta_2 \times$) the frequency resolution of the main (or coarse) SCA to avoid frequency tuning “dead zones” over PVT variations.

During the optimization, we targeted a sub-1.5-kHz frequency resolution with δ_1 and δ_2 set to 120%. The DE algorithm efficiently produced multiple layouts that met the target resolution and design constraints in <40 seconds. To select the optimal layout, we evaluated how the frequency resolution was affected by potential fabrication errors, as on-chip passive components are prone to inaccuracies caused by the lithography, such as variations in line width and coil position. As shown in Fig. 9, leveraging the AI surrogate

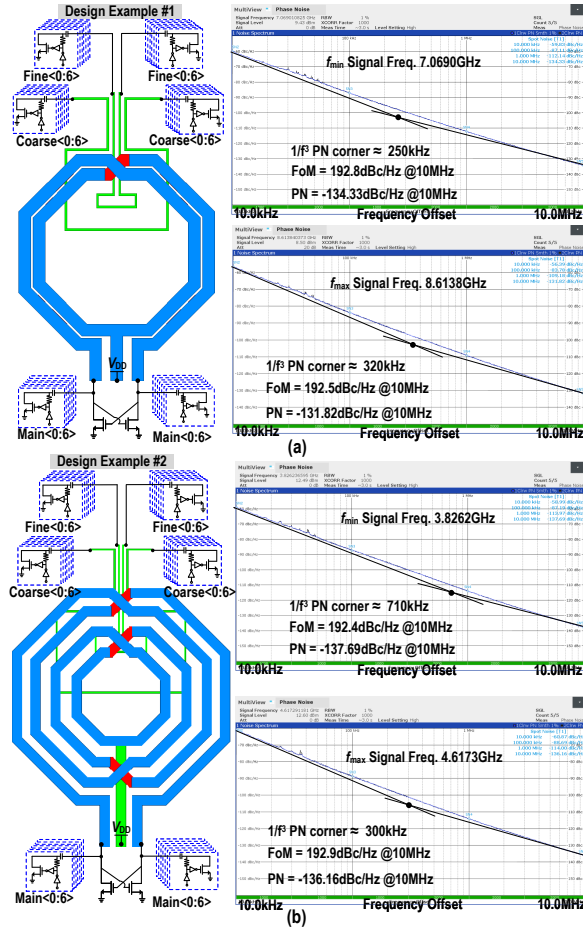


Fig. 11. (a) Schematic of design example #1 with a two-turn inductor as the main tank, along with measured phase noise plots at the minimum and maximum frequencies. (b) Design example #2 with a four-turn inductor as the main tank. The main, coarse, and fine frequency tunings for both designs are realized using three 7-bit SCAs with existing PDK capacitors.

model, we were able to complete 1,024 Monte Carlo simulations in just 15 seconds on a personal laptop, compared with ~ 90 hours using EM simulations on a 64-core workstation. This capability of quickly predicting the impact of process variations is particularly valuable for high-volume production.

IV. EXPERIMENT RESULTS

The proposed AI-assisted DCO design automation was demonstrated using two examples, taped out using the GlobalFoundries 22-nm CMOS SOI process. The die micrographs are shown in Fig. 10.

The first design targets emerging beyond-5G communications in the upper mid-band with a tuning range of 7.1 to 8.6 GHz and employs a two-turn inductor as its main tank [see the schematic in Fig. 11(a)]. The measured frequency tuning resolution is 0.225 to 1.5 kHz, the $1/f^3$ phase noise corner is 250 to 320 kHz, and the $\text{FoM}_{10\text{MHz}}$ is 192.5 to 193.0 dBc/Hz across the whole tuning range. The measured phase noise plots are also shown in Fig. 11(a).

The second design targets a major 5G communication band, the n77 band, with a tuning range of 3.8 to 4.6 GHz. It utilizes

TABLE III
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART DCOs AT
SIMILAR FREQUENCIES

Ref.	ISSCC 2017 [28]	ISSCC 2019 [32]	RFIC 2022 [26]	This Work	
Main Tank Topology	Trifilar-Coil	Folded-Stacked	Class-B	Class-F ₂	
Freq. Tuning Mechanism	XFMR Feedback	SCA	SCA	Weakly Coupled Aux Coils	
Freq. (GHz)	3.2 to 4	4.15 to 4.97	8.2 to 10.2	7.1 to 8.6	3.8 to 4.6
Tuning Range	22%	18%	24%	19.8%	18.8%
Res. (kHz)	1.3	N/A	17.5	<1.5	
V _{DD} (V)	0.4	0.2	1.2	0.4	
PN _{10MHz} (dBc/Hz)	-145 to -143	-144.3 to -142.2	-134° to -129°	-134.3 to -131.8	-137.8 to -136.2
FoM _{10MHz} (dBc/Hz)	190	191 to 195.8	178.0° to 181.3°	192.5 to 193.0	192.4 to 193.3
1/f ³ PN Corner (kHz)	120	70 to 700	N/A	250 to 320	300 to 710
P _{DC} (mW)	3.8	1.77 to 2.28	13	0.63 to 0.71	0.45 to 0.49
Core Area (mm ²)	0.11	0.272	0.13	0.11	0.16
Custom Cap?	Yes	N/A	Yes	No	
Design Automation?	No	No	No	Yes (AI-assisted end-to-end synthesis.)**	
Tech.	16-nm FinFET	22-nm FDSOI	28-nm CMOS	22-nm FDSOI	

* Graphically Estimated.

** Dataset collection: 77 hours, training: 1 hour, inverse design: 80 seconds.
 $\text{FoM} = |\text{PN}| + 20 \log_{10}(f_0/\Delta f) - 10 \log_{10}(P_{\text{DC}}/1\text{mW})$

a four-turn inductor as its main tank [see the schematic in Fig. 11(b)]. The measured frequency tuning resolution is 0.525 to 1.5 kHz, the $1/f^3$ phase noise corner is 300 to 710 kHz, and the $\text{FoM}_{10\text{MHz}}$ is 192.4 to 193.3 dBc/Hz across the tuning range.

Compared with prior works at similar frequencies (see Table III), both designs achieve state-of-the-art performance, with $>192.4\text{-dBc/Hz}$ $\text{FoM}_{10\text{MHz}}$ and $<1.5\text{-kHz}$ frequency resolution. It is important to note that such high performance is achieved in a compact, single-inductor footprint and without using any custom capacitors as in other manually designed DCOs [26], [28]. These results clearly demonstrate the effectiveness of our proposed DCO design automation flow and its capability to achieve state-of-the-art DCO performance that outperforms existing human expert designs.

V. CONCLUSION

In this paper, we present a two-step AI-assisted end-to-end DCO design automation flow and demonstrate its effectiveness using two silicon prototypes. Their measurement results highlight the unique advantages of our proposed design flow, including (i) significantly reducing design time from several weeks (per our own design experience) to ~ 80 seconds (other than one-time 77 hours of data collection and 1 hour of training), (ii) achieving globally optimal designs, (iii) frequency agility (1-20 GHz), and (iv) generating new, unseen passive networks to achieve functions beyond expert designs. This automated design flow can be readily extended to other frequencies, processes, and building blocks, offering a promising solution to enhance RF IC design productivity.

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